Cyber Kill Chain

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Learning Objective

1. Understand the phases of cyberattacks per the Cyber Kill Chain framework.
2. Identify relevant internal data sources for detecting threats and monitoring security events.

Task at Hand

1. The table below lists the seven phases of the cyber kill chain. Informed by the reading and recorded demo, complete columns A, B, and C as follows:

    a. In the **Description** column (**column A**), provide a ***short* description** of each of the 7 phases of the cyber kill chain. Please do not simply copy/paste from the article or any other source. Use your words and based on your own understanding.
    b. In the **Example** column (**column B**), provide **at least** one ***specific* example** from the demo of an activity that I performed with respect to each of the 7 phases.
    c. In the **Evidence** column (**column C**), and for ***each*** example you listed in column B, provide a supporting **screenshot** (or more) captured from the demo. You may list the screenshots **below** the table but make sure to provide a good mapping between the Evidence from the table (e.g., figure numbers) and the corresponding screenshots (e.g., figure numbers plus captions).
    d. Should you choose to use a different format to document your notes other than a Table (i.e. an Outline), that is fine, so long as it is well-formatted (easy to read) and includes the Description, Example, and Evidence items discussed above.

Phase	(A) Description	(B) Example	(C) Evidence
Reconnaissance	Surface level finding on websites for basic public information like usernames and emails.	We were able to find Amazon's IP address, ISP, and even where the servers are located.	Figure 1
Weaponization	A combination of using automated tools and found information to find more sensitive information.	Using Zap, we were able to find that there is possibility of SQL Injection. With SQLmap command we learned that the site uses MYSQL	Figure 2

Phase	(A) Description	(B) Example	(C) Evidence
Delivery	Sending a malicious attachment file disguised as a regular attachment to an unknown user.	After generating the attack file, we logged into Ethan's account to email himself and mass email others with the attached malicious file.	Figure 3
Exploitation	Attacker uses tools like Metasploit to locate vulnerabilities on victim's systems	We used the meterpreter command to gain admin privileges.	Figure4
Installation	Once the victim(s) open the malicious file, it will execute code on the system to further find vulnerabilities to exploit. This would be done without the victim's knowledge.	After Ethan downloaded Putty.exe, we were able to gain access to their local host.	Figure 5
Command & Control (C2)	The attacker establishes a backdoor communication channel to remotely manage the victim's systems to achieve their goals.	Once we gain access to admin privileges, we ran a scheduled task command for nc.exe that will allow us to continue to have access to Ethan's system even after a restart.	Figure 6
Actions on Objectives	Where the attacker is able to execute their goals and/or gaining network access and control.	The attacker's objective was to gain access to Ethan's system and steal his data (login info).	Figure 7

2. Provide at least four[1] different internal data sources (e.g., IDS logs) that Hackazon should have in place to detect the compromise.

- Firewall/Network Logs – It would help identify suspicious outbound connections to the the command and control servers.
- Endpoint Detection and Responses (EDRs) – It would alert suspicious privilege escalation attempts.
- Webserver Logs – They would record HTTP requests, errors, and any suspicious input patterns.
- Account Logs – These would track and detect the failed login attempts and privilege changes during the exploitation process.

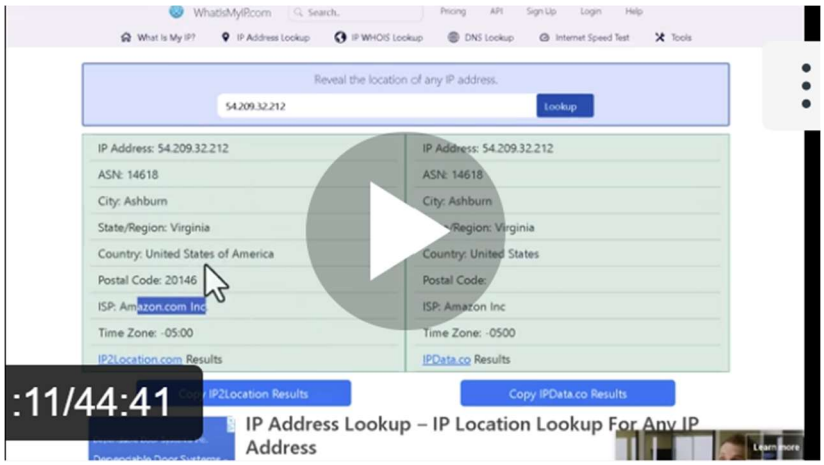


Figure 1 Example of attack on the reconnaissance level.

[1] Other than the IDS logs data source that I provided.

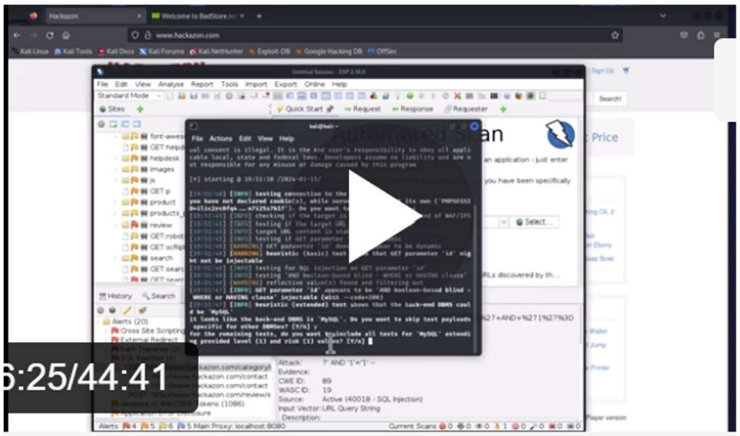


Figure 2 Example of attack on the Weaponization level.

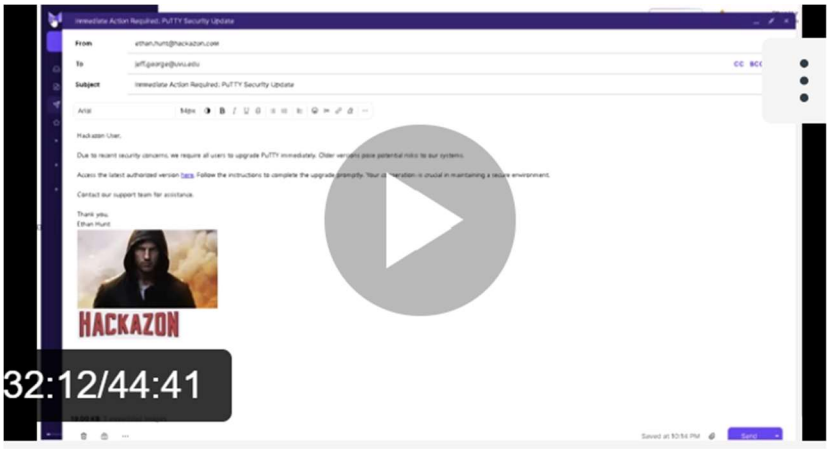


Figure 3 Example of attack in the Delivery level.



Figure 4 Example of backdoor code attack on the Exploitation level.

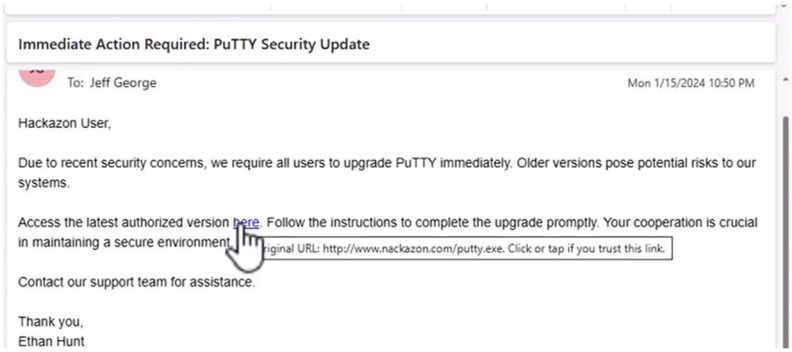


Figure 5 Example of Ethan clicking on malicious file with putty.exe installation.

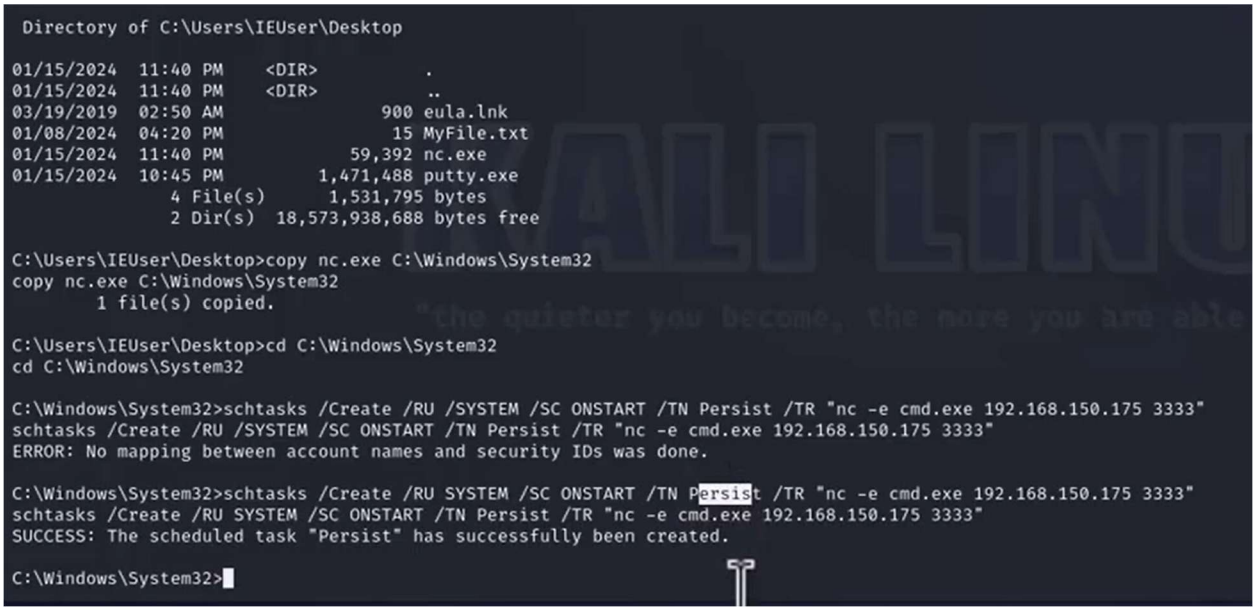


Figure 6 Example of attacker creating a backdoor communication on the victim's system.

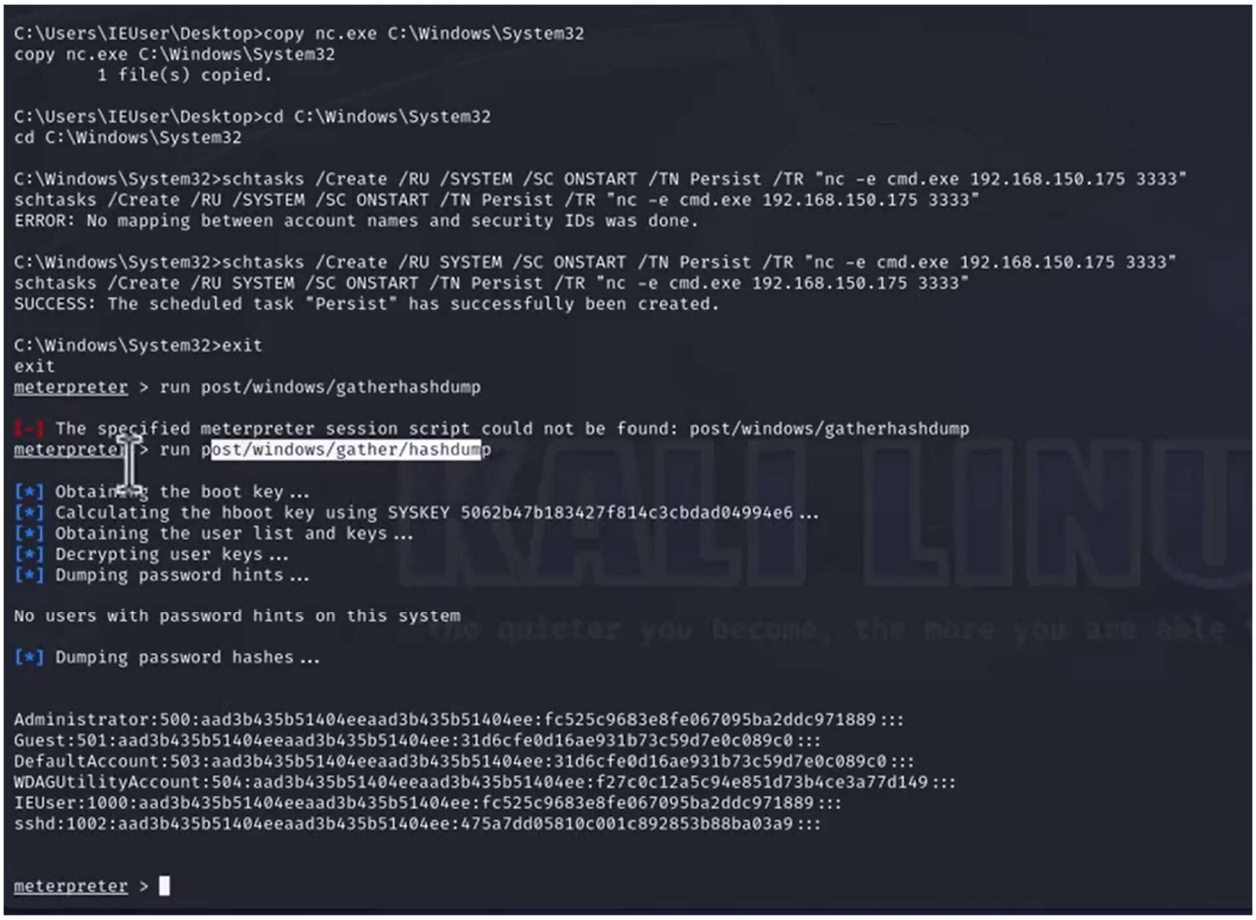


Figure 7 Attacker achieving their objective of stealing data.